# LOT Computer ??? Hardware Manuals

# LOT COMPUTER (Hardware) — Plan, Feasibility & Roadmap

Doc ID: LOT-HW-01 · Rev A · 2026-10-20 · Owner: Vadik Marmeladov (Inventor, COSMO® CIA) Status: PLAN — not yet validated on hardware. All prices are estimates and must be re-quoted.

## 1. Product in one line

A 40 mm stainless-steel tile: a polished mirror face on one side; camera, small screen and one button on the other. It is a pager for lot-systems.com. The site sends autonomous notifications ("Coffee time!"), the screen shows them, and the button ("Copy") sends a signal back to the site's **Log** tab.

## 2. Requirement trace (your 19 points)

| # | Requirement | Where it is handled | Status |
|---|---|---|---|
| 1 | PCBWay | PCB fab + SMT assembly + 100-unit run, see 02-BOM §6 | Planned |
| 2 | Pager-like notifications from AI site | 03-FIRMWARE-API-SPEC §3 | Spec only; server endpoints not built |
| 3 | 2-part stainless body | 04-MECHANICAL §2 | Planned |
| 4 | Flat silver square 4×4 cm × 5 mm | 04-MECHANICAL §1. **5 mm is not feasible with all parts (see §3 risk R1). v1 = 40×40×9 mm.** | RISK |
| 5 | Camera | OV2640 on ESP32-S3 | Planned |
| 6 | LOT API connector | 03-FIRMWARE-API-SPEC §2 | Spec only |
| 7 | PDF manuals | 05-MANUALS (+ `build-pdf.sh`) | PDF built this session |
| 8 | Compress info each session | 06-SESSION-COMPRESSION | Process defined |
| 9 | Firmware documents | 03-FIRMWARE-API-SPEC | Done (draft) |
| 10 | Software to connect with firmware | `tools/` plan (provisioning CLI, §6) | Spec only |
| 11 | Separate documents | This folder: 01…06 | Done |
| 12 | Charger | Qi pad (Qi receiver in device) | Planned |
| 13 | 100 units run | Roadmap Phase 5 | Planned |
| 14 | Weather sensor | BME280 / BME688 | Planned |
| 15 | AI-grade off-the-shelf sensors | BME688 (BSEC gas AI), SGP41, optional IMU | Planned |
| 16 | Button "Copy" → Log tab | Firmware event `device_copy`, spec §2.3 | Spec only |
| 17 | One side polished stainless | Mirror (#8 / BA) finish, 04 §3 | Planned |
| 18 | Other side camera + screen + button | 04 §1 | Planned |
| 19 | Wireless charger | Qi RX, 02-BOM; **steel conflicts with Qi, see R2** | RISK |

## 3. Top risks (read first)

- **R1 — 5 mm thickness.** Stack: LiPo ≥ 2.5 mm, display module ≈ 2.0 mm, PCB 0.8 mm, ESP32-S3 module 2.4 mm, camera 4–5 mm, Qi coil ≥ 0.6 mm, steel walls 2 × 0.5 mm. That is roughly 9–11 mm. Options: (a) 40×40×9–10 mm v1, with camera bump flush; (b) 5 mm only if the camera is dropped, or the unit is split: 5 mm tile + separate battery/charger puck; (c) read item 4 as the charger pad (40×40×5 mm silver square). That reading is workable, and the plan supports it (see 04 §5). **Decision needed from Vadik.**
- **R2 — Stainless blocks Qi.** Conductive steel absorbs the field (eddy currents, heat). Fix: a non-metal back window (ceramic/PEEK/glass, ≥ 25 mm) over the coil, with a ferrite shield layer behind the coil. A fully closed steel shell will not charge.
- **R3 — Stainless blocks Wi-Fi/BLE.** Antenna must sit behind a plastic/glass window or use an external ceramic antenna at a non-metal edge gap. The 2-part body gives a natural seam with a non-conductive gasket.
- **R4 — Privacy (camera).** Always-on camera in a "pager" is sensitive. v1: camera is captured only on explicit press-hold, with a hardware LED and shutter indicator, and no continuous streaming. Required by design.
- **R5 — Site endpoints do not exist yet.** The repo has user-session cookie auth only (`src/server/routes/auth.ts`); no device tokens. See spec §1.
- **R6 — Sites unreachable in this run.** brand/about/institute pages returned DNS errors from the sandbox, so brand tokens (colors, logo rules, CQGS) were NOT applied. Re-run with access.

## 4. Architecture

```
lot-systems.com (AI engine, Log tab)
       ▲ HTTPS/TLS (device token)       ▼ notification (poll + optional push)
  ESP32-S3 (Wi-Fi) ── SPI ── display 1.54" IPS 240×240
       ├── DVP ── OV2640 camera
       ├── I²C ── BME688 (T/H/P/gas) , optional IMU
       ├── GPIO ─ button "Copy"
       └── PMIC ─ Qi RX → LiPo 3.7 V
```

Chosen MCU: **ESP32-S3-WROOM-1-N16R8** (Wi-Fi, BLE 5, native camera DVP, PSRAM, USB-C for flashing, ESP-IDF OTA).

## 5. Roadmap (calendar assumes start 2026-11-01)

| Phase | Weeks | Deliverable | Exit gate |
|---|---|---|---|
| 0 Decisions | 1 | Resolve R1, R2, brand pull | Dimensions frozen |
| 1 Dev-kit prototype | 2–5 | ESP32-S3 dev board + breakout parts, firmware talks to a staging API | "Coffee time!" shows; button creates Log entry |
| 2 Server connector | 3–6 | Device token, `/api/device/*`, Log event `device_copy` visible in Log tab | Server tests green |
| 3 Custom PCB v1 | 5–10 | Schematic, layout, PCBWay 5-pc proto | Boots, charges on Qi, camera frame |
| 4 Enclosure | 6–12 | Two-part steel prototypes (CNC/ MIM quote) | Fit + Qi window + RF test |
| 5 EVT/DVT | 12–18 | 20 units, drop/thermal/Qi/RF tests | Pass list in 05 |
| 6 Pilot run | 18–26 | **100 units** via PCBWay assembly + body supplier | Yield ≥ 90 %, QC report |
| 7 Ship | 26–30 | Manuals PDF, OTA, support | First 100 delivered |

Critical path: enclosure (steel lead time 6–10 weeks) and Qi-through-steel validation. Start both in Phase 1.

## 6. Software to connect with firmware (`tools/lot-device`)

CLI (Node/TypeScript, matches repo): `lot-device flash`, `provision` (Wi-Fi + token), `log` (serial tail), `test-notify "Coffee time!"`, `ota`. Not yet created.

## 7. Cost model (100 units, estimate, USD, unverified)

See 02-BOM §7: ≈ 65–150 per unit electronics + ≈ 40–90 per unit steel body at 100 qty. Rough total ≈ 18k–35k including tooling/NRE and contingency (02-BOM §7).

## 8. Open questions for Vadik

1. 5 mm: device or charger? (R1)
2. Battery life target (days between charges)?
3. Wi-Fi only, or add cellular (adds antenna + SIM, thicker)?
4. Who gets a device: one device per account? Pairing by QR on Settings page?
5. Brand: engraved logo on mirror face?

# LOT COMPUTER — Bill of Materials & Sourcing (v1, 40×40×~9 mm)

Doc ID: LOT-HW-02 · Rev A · 2026-10-20 **Prices are rough estimates from memory of typical 2025–26 market pricing; not live quotes. Links are manufacturer/distributor search or product pages; verify part availability before ordering.**

## 1. Core electronics

| # | Item | Part (preferred) | Qty/unit | Est. unit $ @100 | Source link |
|---|---|---|---|---|---|
| 1 | MCU module | Espressif ESP32-S3- | 1 | 5.5 | https://www.espressif.com/en/products/modules |

| # | Item | Part (preferred) | Qty/unit | Est. unit $ @100 | Source link |
|---|---|---|---|---|---|
| | | WROOM-1-N16R8 | | | |
| 2 | Display | 1.54" IPS 240×240 ST7789 (≈ 31.5×33 mm) | 1 | 4–7 | https://www.aliexpress.com/w/wholesale-1.54-inch-ST7789.html |
| 3 | Camera | OV2640 2MP DVP module (fits 40 mm; thin FPC) | 1 | 3–5 | https://www.aliexpress.com/w/wholesale-OV2640-camera-module.html |
| 4 | Weather/AI gas | Bosch BME688 (T/H/P + gas, BSEC AI) | 1 | 8–10 | https://www.bosch-sensortec.com/products/environmental-sensors/gas-sensors/bme688/ |
| 5 | Weather (low-cost alt) | Bosch BME280 | alt | 2.5 | https://www.bosch-sensortec.com/products/environmental-sensors/humidity-sensors-bme280/ |
| 6 | VOC/NOx (optional) | Sensirion SGP41 | opt | 7 | https://sensirion.com/products/catalog/SGP41 |
| 7 | IMU (tap/orientation) | Bosch BMI270 or ST LSM6DSO | opt | 2–3 | https://www.bosch-sensortec.com/products/motion-sensors/imus/bmi270/ |
| 8 | Qi receiver IC | TI BQ51013B (5 V, ≤ 1 A) | 1 | 2.5 | https://www.ti.com/product/BQ51013B |
| 9 | Qi RX coil | Würth 760308101208-class (≈ 20 mm, thin) + ferrite sheet | 1 | 2–3 | https://www.we-online.com/en/components/products/WPCC_WIRELESS_POWER_CHARGING_COIL |
| 10 | LiPo charger | TI BQ25185 or Microchip MCP73831 | 1 | 0.8–1.5 | https://www.ti.com/product/BQ25185 |
| 11 | 3.3 V buck-boost/LDO | TPS63802 / AP2112K-3.3 | 1 | 0.5–1.5 | https://www.ti.com/product/TPS63802 |
| 12 | Battery | LiPo ≈ 3.7 V 250–350 mAh, ≤ 3.0 mm thick (e.g., 303040-class) | 1 | 3–4 | https://www.aliexpress.com/w/wholesale-303040-lipo.html |
| 13 | Button | Low-profile tactile 3×3 mm or piezo/capacitive under steel | 1 | 0.1–0.5 | https://www.digikey.com/en/products/filter/tactile-switches/197 |
| 14 | Status LED (camera indicator) | 0402 RGB LED | 1 | 0.1 | https://www.digikey.com |
| 15 | Passives, connectors, USB-C (service pads or 16-pin) | – | – | 2–4 | – |
| 16 | PCB | 4-layer, 0.8 mm, ENIG, ~38×38 mm | 1 | 2–4 | https://www.pcbway.com |
| 17 | SMT assembly | PCBWay Turnkey Assembly | 1 | 6–12 | https://www.pcbway.com/pcb-assembly.html |

Electronics estimate: **~ 55–85 / unit at 100 pcs** parts + ~ 8–16 PCB/assembly = **≈ 65–100**, plus 20–30 % contingency → ≈ 95–150.

## 2. Body & materials

| Item | Spec | Est. $ @100 | Source |
|---|---|---|---|
| Front body (screen/cam/button side) | 316L, 0.8–1.0 mm wall, CNC or deep-draw, openings for window, lens, button | 20–40 | PCBWay CNC: https://www.pcbway.com/cnc-machining-quote.html |
| Back body (polished mirror side) | 316L plate, #8 mirror/BA finish, **Qi window cut-out** | 15–35 | PCBWay or Xometry: https://www.xometry.com |
| Qi window | Zirconia ceramic / PEEK / Gorilla-type glass disk 26 mm × 0.6 mm | 3–8 | custom |
| Display window | Sapphire or chem-strengthened glass 0.5 mm, AR coating | 2–5 | custom |
| Camera lens window | AR-coated glass 6 mm | 1 | custom |
| Gasket + adhesive | silicone/ PSA, IP54 target | 1 | – |
| Screws | M1.2 torx, 4× | 0.6 | – |

Note: **316L** is preferred (corrosion-resistant, mostly non-magnetic); 304 acceptable.

## 3. Charger (separate product)

Qi pad, 5 W, 40×40×5 mm silver (the "flat silver square" under reading (c) in 01 §3 R1). Reference ICs: Renesas P9242/ P9221 TX or off-the-shelf low-profile Qi TX pad 3–5 W (retail 8–12). Alternative for v1: ship a standard Qi pad + USB-C cable (cost ≈ 8–12).

## 4. Dev kit (week 1 purchase, ≈ $120)

- ESP32-S3 camera dev board (e.g., Freenove ESP32-S3-WROOM CAM or Seeed XIAO ESP32S3 Sense) — https://www.seeedstudio.com/XIAO-ESP32S3-Sense-p-5639.html
- 1.54" ST7789 display, BME688 breakout (Adafruit 5046 / Pimoroni), Qi RX module + LiPo, tact button, Qi pad.
- Adafruit BME688 breakout: https://www.adafruit.com/product/5046

## 5. Test equipment

USB-C logic analyzer, multimeter, Qi power-meter (e.g., USB-C meter), 2.4 GHz spectrum check (NanoVNA for antenna tuning).

## 6. PCBWay ordering checklist

1. Upload Gerber + BOM (CSV: designator, MPN, qty) + Pick&Place.
2. Choose: 4-layer, 0.8 mm, ENIG, green/black mask, impedance control not needed (DVP at 24 MHz; keep traces short).
3. Assembly: Turnkey; choose consigned or PCBWay-sourced parts; request DFM review for 0201/ 0402 and BME688 LGA.
4. First order: 5 PCBs assembled (proto); second: 20 (EVT); third: 100 (pilot). Request X-ray for BGA/LGA if any.
5. Keep panel: 2×N array with tabs; fiducials; test pads for pogo-pin programming jig.

## 7. 100-unit cost summary (estimate)

| Line | Low | High |
|---|---|---|
| Electronics + PCB + assembly | 6,500 | 10,000 |
| Steel body (2 parts) + windows | 4,000 | 8,000 |
| Battery, gasket, screws, packaging | 800 | 1,500 |
| Tooling/ fixtures/ NRE | 3,000 | 8,000 |
| Test jig + QC labor | 500 | 1,500 |
| Contingency 20 % | 3,000 | 6,000 |
| **Total (≈)** | **~18,000** | **~35,000** |

Re-quote everything before committing.

# LOT COMPUTER — Firmware & LOT API Connector Spec (draft)

Doc ID: LOT-HW-03 · Rev A · 2026-10-20 Status: **specification only.** None of the `/api/device/*` endpoints exist in the repo yet. Current auth is cookie-session (`src/server/routes/auth.ts`); logs are in the `Log` model (`userId, event, text, metadata, context`), listed via

`GET /api/logs` with a whitelist of displayable events (`src/server/routes/api.ts`).

## 1. Server work needed (small)

1. `Device` model: `id, userId, tokenHash, name, lastSeenAt, fwVersion`.
2. Pairing: signed-in user opens Settings → "Pair device" → 6-digit code (10 min TTL). Device posts the code → receives a long-lived device token (stored hashed).
3. Routes under `/api/device` (prefix registered like `/api/public`), auth by `Authorization: Bearer <device token>`, rate limited.
4. Add events `device_copy`, `device_notification_ack`, `device_telemetry` to the `displayableEvents` whitelist so they appear in the Log tab.
5. Notification queue: AI engine / scheduled jobs write `DeviceNotification {id, userId, text, priority, expiresAt, deliveredAt}`.

## 2. Wire protocol (HTTPS, JSON, TLS 1.2+, pinned CA)

### 2.1 Pair

`POST /api/device/pair {code, hwId, fw} → {token, pollSeconds}`

### 2.2 Poll notifications (pager behaviour)

`GET /api/device/notifications?since=<id> → [{id, text, priority, expiresAt}]` Default poll 60 s while awake; long-poll (`?wait=25`) option. Phase 2: MQTT/WebSocket push if battery allows. Device shows the text (max 64 chars, 2 lines + icon), buzzes LED, and sends ack: `POST /api/device/ack {id}` → Log event `device_notification_ack`.

### 2.3 Button "Copy"

Short press: `POST /api/device/event {type:"copy", notificationId?, ts}` → server writes `Log{event:"device_copy", text: <notification text or current clip>, metadata:{deviceId, fw, battery}}`, visible in the Log tab on lot-systems.com. Screen confirms "COPIED ✓". Offline: queue up to 50 events in flash, flush on reconnect. Long press (1.5 s): capture photo (LED on) → `POST /api/device/photo` (JPEG ≤ 200 KB) → Log `device_photo`. Camera is never enabled without press (R4).

### 2.4 Telemetry

Every 15 min: `POST /api/device/telemetry {temp, rh, pressure, iaq, battery, rssi}` → `device_telemetry` (also feeds the existing weather/context). BME688 IAQ computed on-device with Bosch BSEC.

## 3. Firmware architecture (ESP-IDF 5.x, C/C++)

- Tasks: `net` (Wi-Fi, TLS, queue), `ui` (LVGL, display), `sensor` (I²C), `input` (button debounce, GPIO wake), `power` (light sleep, Qi detect).
- Power: deep sleep between polls (target ≥ 3 days at 60 s wake duty — to be measured; Wi-Fi light sleep with DTIM ≈ 5–10 mA avg is the critical figure).
- Storage: NVS (token, Wi-Fi), SPIFFS (event queue).
- OTA: dual-partition, signed images (secure boot v2, flash encryption), pulled from `GET /api/device/firmware`.
- Security: per-device token, no secrets in firmware image, TLS cert pinning, camera opt-in flag server-side.
- Provisioning: BLE or USB serial via `lot-device provision` (see 01 §6).

## 4. Screen UX (240×240)

Idle: dark, clock + temp. Notification: large text, e.g. **Coffee time!**, soft pulse; button = Copy; second press dismisses. Colors/typography to be taken from brand.lot-systems.com (unavailable this run).

## 5. Test plan

Unit: protocol mock server. HIL: button → Log entry in staging DB. Soak: 72 h poll/ack. RF: with steel body + window. Power: measure µA sleep.

# LOT COMPUTER — Mechanical & Industrial Design

Doc ID: LOT-HW-04 · Rev A · 2026-10-20

## 1. Envelope

Target 40 × 40 mm. Height: **5 mm requested; v1 baseline 9 mm** (see 01 §3 R1). Corner radius 3 mm, edge chamfer 0.3 mm.

Stack (v1, mm): back plate 0.8 · ferrite+coil 0.7 · battery 3.0 · PCB+components 2.0 (components both sides, 0.8 board) · display/cam layer · front plate 0.8 ≈ 9.

## 2. Two-part body

- **Part A (back):** polished 316L, mirror face outward. Centre Ø26 mm ceramic/PEEK Qi window flush-bonded (R2). Inner ledge for coil + ferrite.
- **Part B (front):** 316L frame with: 31 × 33 display window, Ø6 camera window, button hole (or capacitive zone), 2 mm antenna gap at one edge with plastic insert (R3).
- Joint: 4 × M1.2 screws or laser-weld seam (non-serviceable). Silicone gasket → IP54.

## 3. Finish

Polished side: mirror (#8). Other side: brushed/bead-blasted to avoid glare and fingerprints on camera side. Passivation per ASTM A967.

## 4. Manufacturing

Prototype: CNC (PCBWay/Xometry). 100 pcs: CNC still OK ($); >1000: MIM/stamping. Polishing is a hand-labor cost driver; spec the mirror grade and inspect under 10× light.

## 5. Charger reading (c): 40×40×5 mm silver Qi pad

Aluminum-free top (PC/ceramic) over TX coil; stainless or anodized aluminum frame ring; magnets optional (check against compass/IMU). Max 5 W.

## 6. Tests

Drop 1 m ×6 faces, 85 °C/ 85 % RH, Qi efficiency through window (target ≥ 60 %), thermal rise ≤ 10 °C during charge, RF throughput at −70 dBm.

# LOT COMPUTER — User Manual (draft)

Doc ID: LOT-HW-05 · Rev A · 2026-10-20 · Draft; features marked ★ depend on unbuilt server endpoints.

## 1. In the box

LOT Computer, Qi charging pad, USB-C cable, quick-start card.

## 2. First start

1. Charge: place the polished side *down* on the pad (the charging window is on the mirror side). Full charge ≈ 2 h.
2. On lot-systems.com open Settings → Pair device ★ and read the 6-digit code.
3. Hold the button 5 s; the screen shows "Pair". Enter the code via the setup page or the `lot-device` tool.

## 3. Daily use

- A message appears, e.g. **Coffee time!** The LED pulses once.
- Press the button once: **Copy** — the text is logged to the Log tab ★.
- Press again: dismiss.
- Hold 1.5 s: takes a photo (LED on) ★.

## 4. Care & safety

Do not use near strong magnets for first-aid devices. Do not submerge (IP54). Battery: Li-polymer — do not puncture or heat above 60 °C. FCC/CE testing is required before sale (not yet done).

## 5. Troubleshooting

No notifications: check Wi-Fi, re-pair. Not charging: ensure the mirror side is on the pad, remove metal objects between.

## 6. Privacy

Camera only runs on long-press. Photos upload only to your LOT account.

# Session compression protocol (point 8)

At the end of each session, append ≤ 15 lines to `docs/hardware/STATE.md`: `DECISIONS · OPEN QUESTIONS · NEXT 3 ACTIONS · RISKS CHANGED · NUMBERS CHANGED`. Each new session starts by reading STATE.md only (not the full docs). Full report goes to `docs/LOT-SR-<date>-HW<nn>.md`.